(BT102)

→ From the time of Aristotle, living organisms were placed into just two groups: Plants & Animals.

→ Carl Linnaeus in 1735 proposed first classification system with two kingdoms: Plantae & Animalia.

→ 1857 → Carl von Naseli proposed bacteria & fungi be placed in kingdom Plantae.

→ 1866 → Ernst Haeckel placed bacteria protozoa fungi & algae in Protista.

→ 1937 → Edouard Chatton introduced prokaryotes.

→ 1959 → Fungi were placed in kingdom Fungi.

→ 1968 → Robert Murray proposed kingdom Prokaryotae for bacteria.

→ 1969 → five kingdom classification by Robert Whittaker.

→ 1977 → Carl Wese proposed three domain classification.

→ System of scientific nomenclature

↓

Carl Linnaeus

→ 1668 → Italian scientist, Francesco Redi challenged spontaneous generation theory.

→ 1745 → John Needham reported that microbes can rise spontaneously from heated nutrients fluids when cooled & sealed in flasks.

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→ 1765 → Italian scientist, Lazzaro Spallanzani proposed that organism in broth prepared by John Needham may have entered from air after they were boiled & cooled before putting them in flask.

→ Importance of oxygen to life.

↓

Anton Laureit Lavoiser

→ 1861 → Louis Pasteur resolved issue of spontaneous generation.

→ Golden age of microbiology

↓

1857 – 1914

→ 1865 → Pasteur established causative agent for silkworm disease.

→ Joseph Lister → started using disinfectant for the first time (Listerine)

→ 1876 → Robert Koch proved that bacterium (Bacillus anthracis) causes disease in animals.

→ 1880 → Pasteur discovered vaccination

→ 1796 → Edward Jenner used similar technique of vaccination for small pox.

→ 1910 → Paul Ehrlich made the first chemotherapeutic agent salvarsan for treating human against syphilis.

• Sulfonamide was synthesized

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from penicillium chrysogenum

→ 1928 → Alexander Fleming discoved pencilin

Total magnification = Magnification of objective lens x Magnification of ocular lens.

Low = 10X

High = 40X

Oil immersion = 100X.

Ocular lens = 10X

Low Magnification power = 10x10 = 100

Oil immersion power = 100x10 = 1000

→ Resolving power of light microscope = 0.2 µm or 200nm

n = refractive index, 1.5 for immersion oil

Resolution ← $d = \frac{0.5\,\lambda}{n \sin\theta}$

→ uv lamp → Mercury lamp.

→ Resolution of EM = 0.1nm.

Fixing a slide.

↓

→ Heating → Physical method

Chemical method → Methyl alcohol / Ethanol or formaldehyde.

→ Basic dye → positive ion.

→ Acidic dye → Negative ion.

→ Bacteria is slightly negative at pH 7.

↓ Basic dyes are used.

(Basic Dyes)

→ Crystal violet

→ Methylene Blue

→ Malachite green.

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→ Safranin

(Acidic Dyes)

→ Eosin
→ Acid fuchsin
→ Nigrosin

one stain is used ← (Simple dyes) / Methylene blue staining

→ methylene blue
→ carbolfuchsin
→ Crystal violet
→ safranin

couple of stains are used ← (Differential staining (Gram's staining)) → (spores can be seen)

↓

1884 by Hans Christian Gram

← (Special staining) / Endospore staining

↓

Acid Fast staining

1) carbol fuchsin
2) Acid alcohol
3) Methylene blue

Myobacteria

↓

gram positive (red)

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Bacteria

- Prokaryotes
- single celled
- cell wall (peptidoglycan)
- True bacteria

Archea

- Prokaryotes
- lacks peptidoglycan (may have special lipids)
- vary in rRNA sequence from Eubacteria
- live in extreme conditions
- Donot cause disease

Protista

- algae; protozoa, slime molds, water molds
- Eukaryotes
- cell wall (cellulose)
- Photosynthetic & produce oxygen
- unicellular & may be multicellular
  E.g.: Seaweeds & pond scum

Protozoa:-

- Eukaryotes
- Unicellular
- No cell wall
- move by pseudopods, flagella or cilia
- photosynthetic

Slime molds:-

- Fungus like
- protozoan like
- single celled
- aggregate to form reproductive structures

Fungi

- → unicellular
- → may be multicellular
- → cell wall (chitin)

VIROIDS:-

→ composed of circular ss RNA

VIRUSOIDS:-

→ contain circular ss RNA
↓ helps in replication & encapsidation
→ satellite viruses

Water molds:-

→ grows on surface of fresh water & humid soils
→ Also known as mildews & white rusts.
→ look like fungus

Viruses:-

→ acellular structures
↓
protein capsid
↓
enclosing DNA/RNA
↓
sometimes contain extra membrane called Envelop

PRIONS:

→ infectious protein formed inside body
→ act as organism that transmit from person to person
→ resistant to heat

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Staphylococcus aureus

↓

Staphyla → clustered

Cocci → spherical

aureus → gold-colored colonies

E. coli

↓

large intestine

S. aureus

↓

on skin

→ From the time of Aristotle, it was thought that living organism arise from non living matter

Pasteurella multocida → bacterium

↓

fowl cholera in chickens

Blue light → shortest wavelength

Flurochromes → absorbs short wavelength & emit longer wavelength

↓

attached to the antibodies

Electron beam → shorter wavelength

Stains

↓

salts (positive & negative ions)

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Simple Staining

- Stain is washed off
- Mordaal is added

Differential S

- reacts differently with different kinds of bacteria (primary S)
- Crystal violet is added
- crystal violet is washed off
- Iodine (mordant) added (enhance binding)
- washing of slide (alcohol) Decolorizing agent
- Gram + (retain crystal violet)
- Gram - (appear colourless)
- Rinsing (alcohol)
- Staining (safranin) counter stain
- Washing (water)
- Blotted dry & examined

Special S

- Acid Fast (1) Stainings: (Lipophilic) (Red) ↳ mycobacteria (gram +) ↓ waxy material (mycolic acid) in their cell wall
- Mycobacterium tuberculosis ↓ causative agent for tuberculosis
- M. leprae ↓ Leprosy.

Procedure:

→ carbol fuchsin (primary S)
→ Acid alcohol (decolorizer)
→ Methylene blue (counter stain)

Endospores (2) Staining:

→ malachite green (P. S)
→ Heat (steaming for 5 min)
→ wash (water)
→ safranin (C. S)
→ (Red) → vegetative cells)
→ (Green) → spores.

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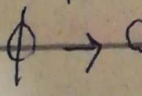
Prokaryotes	Eukaryotes
→ DNA is not enclosed in nuclear membrane	→ DNA → enclosed in nuclear membrane
→ Chromosome → single circular	→ Chromosome → linear, thread like structures
→ DNA → not associated with histones	→ DNA → associated with histones found in multiple chromosomes
→ Membrane bound organelles → not present	→ cell wall → chitin, cellulose
→ cell wall → peptidoglycan complex carbohydrate	→ Membrane bound organelles

Shapes of Bacteria

Coccus

→ circular or spherical

→ Diplococci (division in one plane)



→ streptococci



→ tetrads (division in two plane)



→ sarcinae (division in three plane)

→ staphylococci (division in multiple plane)

Bacillus

→ Rod shaped

→ bacillus (single rod)

→ diplobacilli (Two rods attached)

→ streptobacilli (bacilli arranged one after one)

→ Coccobacillus (cell is neither bacillus nor coccus)

↓

intermediate b/w two extremes

Spiral

→ curved shaped

↓

subgroups

↓

1) Vibrio → curved rod

2) Spirillum → Helical but rigid

3) Spirochete:
- Helical but flexible
- move by axial filament

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(Structures outside cell wall)

Glycocalyx

- outermost layer
- viscous & gelatinous
- consist of polysaccharide & polypeptide
- **Capsule:**
  If glycocalyx is firmly attached to cell.

∴ play role in:
- ⇒ virulence (degree of pathogenicity)
- ⇒ part of vaccine
- ⇒ Negative staining
- ⇒ glistering colonies
- ⇒ E.g.: Bacillus anthraci, streptococcus pneumoniae, klebsiella
- Thickness depends on cultural conditions
- water soluble
- make broth viscous & stringy.

Staining:
- prepare smear
- Fix (acid alcohol) 15 sec
- wash (dH_2O)
- cover (acid fuchsin 1 min)

Flagella (for motility)

- long filamentous structures
- propels bacteria

Classifications:
1) Atrichous → No flagellum
2) Peritrichous → Distributed over entire cell
3) Polar
   - Monotrichous
   - Amphitrichous
   - Lophotrichous

Parts:
1. **Filament:**
   - protein (flagellin) → make H antigen → identification of Gram- bacteria.
2. **Hook:** A different protein
3. **Basal Body:** most complex
   - 4 rings
   - Gram- cells → Two plasma membrane (outer, inner)
   - Gram+ cells → 2 rings in basal body + 1 plasma membrane + thick peptidoglycan

→ bacterium (red)

Capsule (colourless)

→ EPS (extra cellular polymeric substance)

→ Slime:

If glycocalyx is loosely attached.

Movement:

Bacteria → Run & Tumble

Flagella → run & swim (counter clockwise) 1)

→ clockwise (Tumbling)

Stainings:

2)

→ Take a drope of fresh culture

→ Air dry (Do not fix) 3)

→ cover (mordant) → for 10-30 min tannic acid ↓ 10% in 5% NaCl

→ Rinse (water)

→ Add carbol fuchsin (5-15 min)

→ Rinse (water)

→ Air dry

→ see under microscope

Pasteurization:

→ 63°C for 30 min

→ High temperature short time → 72°C for 15 sec

→ ultra-high temperature → 140°C for 4 sec

→ Refrigeration → 2-8°C

Clean object

↓

sterilized

↓

5-8 hours (38°C) or 3-4 hours (54°C)

Humidity → 40-50%

ETO → 700 mg/L

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Microbes

10°C - 110°C

1) Psychrophiles

↓

Strict psychrophiles - 15°C (cold loving)

Psychotrophs - 20-30°C (food spoilage bacteria)

2) Mesophiles - 25°C - 40°C (moderate temp. loving)

↓

37°C (optm - temp)

3) Thermophiles - 50° - 60°C (Heat loving)

↓

hyperthermophiles - 80°C

(Min - Mix)

↓

30°C apart

Food prevention from spoilage → below 5°C or above 60°C

Acidity → pH < 7

Alkalinity → pH > 7.

Ph

organisms → 6.5 & 7.5

few pH below 4

Molds & yeast → pH 5-6.

Obligate halophiles → 30% salt

Facultative → 2% salt

Nitrogen + Sulfur

↓

protein synthesis

N + P

↓ DNA / RNA synthesis & ATP

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NADH + $FADH_2$

↓

36 or 38 ATP molecules

Agar

↓

liquifies — 95°C

solidifies — 43°C

Bacterial Colonies

↓

1) Punctiform
2) Circular
3) Filamentous
4) Irregular
5) Rhizoid
6) Spindle shaped

Elevations:-

1) Flat
2) Raised
3) Convex
4) Pulvinate
5) Umbonate

Margins:-

1) Entire
2) Undulate
3) Lobate
4) Erose
5) Filamenteous
6) Curled

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